

RAINFALL AT Manchester Fern Bank Warrington

70

Lat.

Lon.

County of Lancashire

Rain Gauge, by

Observer

J. King junr Esq

Gauge No I

Diameter.				5							
Height.				1.							
Altitude.				110							
Year.	1870	1871	1872	1873	1874	1875	1876	1877	1878	1879	MEANS.
January.				# 2.32	3.02	3.51	1.46	4.47	3.01	1.53	
February.				.38	1.80	.60	3.86	3.99	1.15	2.72	
March.				2.29	2.83	.56	3.46	2.16	1.56	1.38	
April.				.75	.83	.78	2.86	1.58	1.53	2.11	
May.				1.79	1.21	1.63	1.19	3.20	3.62	1.98	
June.				2.41	.98	2.94	3.66	1.99	2.56	4.94	
July.				3.68	1.60	4.45	2.82	4.97	1.42	3.32	
August.				4.17	3.61	4.11	2.63	4.94	4.88	5.24	
September.				2.11	3.29	4.66	3.74	2.42	4.36	2.90	
October.				3.59	2.97	4.47	2.40	3.63	3.23	2.08	
November.				2.17	4.12	3.62	3.10	4.65	2.23	.94	
December.				.67	2.39	.93	4.01	3.03	2.18	2.01	
TOTALS.				26.33	28.65	32.26	35.19	41.03	31.73	31.15	

40 feet NW of House 4 miles S. of Manchester
 1873 Jan'y one day estimated gauge not being fixed, Oct.
 one day + Nov. one day estimated g. blown over.

70

RAINFALL AT Manchester Fern Bank Withington

Lat. Lon. County of Lancashire

Rain Gauge, by

Observer

J. King Junr Esq.

Gauge No II

Diameter.					5						
Height.					1	-					
Altitude.											
Year.	1870	1871	1872	1873	1874	1875	1876	1877	1878	1879	MEANS.
January.					3.00	3.48	1.47	4.49	3.08	1.51	
February.					1.77	.60	3.74	3.96	1.16	2.78	
March.					2.80	.58	3.48	2.15	1.53	1.39	
April.					.83	.80	2.81	1.54	1.49	2.08	
May.					1.20	1.62	1.16	3.11	3.60	1.99	
June.					1.01	2.91	3.63	1.97	2.49	4.84	
July.					1.60	4.41	2.80	4.91	1.41	3.26	
August.					3.55	4.09	2.59	4.92	4.87	5.15	
September.					3.30	4.63	3.71	2.41	4.32	2.86	
October.					2.95	4.46	2.42	3.69	3.27	2.11	
November.					4.11	3.65	3.13	4.68	2.22	.94	
December.					2.46	.94	4.09	3.08	2.21	2.02	
TOTALS.					28.58	32.17	35.03	40.91	31.65	30.93	

Gauge 100ft. NW of Stouee

Withington Church ½ mile NE.

RAINFALL AT *Manchester Withington*

80

Lat. _____

Lon. _____

County of _____

Lancashire

No. *1* Rain Gauge, by _____

Observer *A. Ring, Esq., Esq.*

Mersey - Upper H Daily

Diameter	<i>5"</i>										
Height	<i>1.0</i>										
Altitude	<i>110 T</i>										
Year	1880	1881	1882	1883	1884	1885	1886	1887	1888	1889	MEANS.
January	<i>60</i>	<i>46</i>	<i>258</i>	<i>3.80</i>	<i>4.16</i>	<i>1.56</i>	<i>4.54</i>	<i>1.97</i>			
February	<i>2.09</i>	<i>4.74</i>	<i>2.46</i>	<i>1.73</i>	<i>1.84</i>	<i>2.21</i>	<i>.70</i>	<i>.40</i>			
March	<i>2.14</i>	<i>3.96</i>	<i>2.04</i>	<i>1.06</i>	<i>2.12</i>	<i>2.09</i>	<i>1.93</i>	<i>1.54</i>			
April	<i>1.17</i>	<i>1.36</i>	<i>3.67</i>	<i>1.77</i>	<i>.75</i>	<i>1.21</i>	<i>1.60</i>	<i>1.69</i>			
May	<i>2.52</i>	<i>3.24</i>	<i>1.36</i>	<i>.81</i>	<i>1.47</i>	<i>2.26</i>	<i>5.84</i>	<i>2.40</i>			
June	<i>2.54</i>	<i>1.99</i>	<i>6.33</i>	<i>3.41</i>	<i>.76</i>	<i>3.45</i>	<i>1.41</i>	<i>1.15</i>			
July	<i>5.76</i>	<i>2.42</i>	<i>5.20</i>	<i>3.22</i>	<i>3.85</i>	<i>1.16</i>	<i>3.62</i>	<i>.76</i>			
August	<i>2.81</i>	<i>4.85</i>	<i>2.65</i>	<i>2.58</i>	<i>1.82</i>	<i>2.17</i>	<i>1.44</i>	<i>1.72</i>			
September	<i>4.22</i>	<i>2.33</i>	<i>2.23</i>	<i>4.66</i>	<i>2.90</i>	<i>4.15</i>	<i>3.40</i>	<i>3.55</i>			
October	<i>2.62</i>	<i>2.41</i>	<i>3.21</i>	<i>4.73</i>	<i>2.21</i>	<i>5.40</i>	<i>5.52</i>	<i>2.12</i>			
November	<i>3.21</i>	<i>3.12</i>	<i>5.14</i>	<i>3.27</i>	<i>1.74</i>	<i>2.66</i>	<i>2.18</i>	<i>.93</i>			
December	<i>4.97</i>	<i>3.73</i>	<i>4.83</i>	<i>4.37</i>	<i>3.49</i>	<i>2.13</i>	<i>3.96</i>	<i>2.41</i>			
TOTALS	<i>34.65</i>	<i>34.61</i>	<i>41.70</i>	<i>33.41</i>	<i>27.11</i>	<i>30.45</i>	<i>36.14</i>	<i>20.64</i>			

Dwelling House 30ft high 75ft

RAINFALL AT *Manchester, Withington* 80

Lat. _____ Lon. _____ County of *Lancashire*
 Rain Gauge, by _____ Observer *L. King, Am., Esq.*

Mersey - Upper H. *Daily*

Diameter	<i>5</i>											
Height	<i>1.0</i>									<i>1.3</i>		
Altitude	<i>110 T</i>											
Year	1880	1881	1882	1883	1884	1885	1886	1887	1888	1889	MEANS.	
January	<i>.59</i>	<i>.46</i>	<i>2.58</i>	<i>3.85</i>	<i>4.13</i>	<i>1.61</i>	<i>4.67</i>	<i>2.07</i>	<i>1.46</i>	<i>1.09</i>	<i>2.251</i>	
February	<i>2.10</i>	<i>4.65</i>	<i>2.49</i>	<i>1.78</i>	<i>1.84</i>	<i>2.34</i>	<i>.71</i>	<i>.40</i>	<i>1.00</i>	<i>1.88</i>	<i>1.919</i>	
March	<i>2.15</i>	<i>3.97</i>	<i>2.07</i>	<i>1.05</i>	<i>2.10</i>	<i>2.07</i>	<i>1.97</i>	<i>1.52</i>	<i>1.63</i>	<i>3.12</i>	<i>2.165</i>	
April	<i>1.17</i>	<i>1.36</i>	<i>3.69</i>	<i>1.74</i>	<i>.70</i>	<i>1.23</i>	<i>1.59</i>	<i>1.63</i>	<i>1.47</i>	<i>2.10</i>	<i>1.668</i>	
May	<i>2.52</i>	<i>3.28</i>	<i>1.34</i>	<i>.80</i>	<i>1.46</i>	<i>2.24</i>	<i>5.91</i>	<i>2.46</i>	<i>.88</i>	<i>2.29</i>	<i>2.818</i>	
June	<i>2.53</i>	<i>2.03</i>	<i>6.32</i>	<i>3.38</i>	<i>.74</i>	<i>3.44</i>	<i>1.49</i>	<i>1.16</i>	<i>3.10</i>	<i>.59</i>	<i>2.478</i>	
July	<i>5.87</i>	<i>2.42</i>	<i>5.22</i>	<i>3.18</i>	<i>3.86</i>	<i>1.15</i>	<i>3.66</i>	<i>.77</i>	<i>5.31</i>	<i>3.09</i>	<i>3.453</i>	
August	<i>2.82</i>	<i>4.84</i>	<i>2.63</i>	<i>2.50</i>	<i>1.82</i>	<i>2.14</i>	<i>1.41</i>	<i>1.69</i>	<i>3.15</i>	<i>5.70</i>	<i>2.870</i>	
September	<i>4.20</i>	<i>2.35</i>	<i>2.23</i>	<i>4.63</i>	<i>2.93</i>	<i>4.16</i>	<i>3.37</i>	<i>3.57</i>	<i>.91</i>	<i>2.52</i>	<i>3.087</i>	
October	<i>2.57</i>	<i>2.41</i>	<i>3.22</i>	<i>4.64</i>	<i>2.25</i>	<i>5.47</i>	<i>5.35</i>	<i>2.15</i>	<i>1.50</i>	<i>2.91</i>	<i>3.268</i>	
November	<i>3.32</i>	<i>3.30</i>	<i>5.20</i>	<i>3.32</i>	<i>1.68</i>	<i>2.80</i>	<i>2.20</i>	<i>.90</i>	<i>4.42</i>	<i>2.04</i>	<i>2.918</i>	
December	<i>5.02</i>	<i>3.71</i>	<i>4.82</i>	<i>2.37</i>	<i>3.54</i>	<i>2.20</i>	<i>4.17</i>	<i>2.46</i>	<i>2.17</i>	<i>2.23</i>	<i>3.269</i>	
TOTALS	<i>34.86</i>	<i>34.78</i>	<i>41.81</i>	<i>33.24</i>	<i>27.05</i>	<i>30.85</i>	<i>36.70</i>	<i>20.79</i>	<i>27.00</i>	<i>29.56</i>	<i>31.664</i>	

Dwelling House 30ft high 40ft
Withington Church 1/4 mile NE

RAINFALL AT Manchester Withington

80

Lat.

Lon.

County of

Lancashire

Snowdon Rain Gauge, by Casella

Observer

J. King Jun, Esq

Mersey. Upper H.

Diameter			8								
Height			1								
Altitude			110 T								
Year	1880	1881	1882	1883	1884	1885	1886	1887	1888	1889	MEANS.
January... ..			# 2.54	3.75	4.07	1.57	4.54	2.05	1.43	1.08	
February			2.42	1.77	1.82	2.31	.70	.39	.97	1.87	
March			2.03	1.06	2.06	2.05	1.91	1.49	1.57	3.05	
April			3.64	1.71	.72	1.21	1.58	1.63	1.45	2.09	
May			1.31	.79	1.43	2.22	5.72	2.39	.91	2.27	
June			6.20	3.34	.75	3.37	1.44	1.12	3.07	.58	
July			5.09	3.12	3.81	1.13	3.55	.75	5.28	3.09	
August			2.59	2.46	1.77	2.09	1.40	1.69	3.10	5.61	
September....			2.17	4.88	2.87	4.08	3.28	3.51	.91	2.49	
October			3.12	4.89	2.19	5.37	5.39	2.13	1.55	2.87	
November...			5.02	3.24	1.68	2.71	2.14	.91	4.41	2.04	
December			4.76	2.26	3.50	2.08	4.08	2.37	2.12	2.22	
TOTALS			40.89	32.67	26.67	30.19	35.73	20.43	26.77	29.26	

*Withington Church $\frac{1}{2}$ mile NE
Dwelling House 30ft high 20^{yard} E*

RAINFALL AT *Manchester Withington No 2*

Lat. _____

Lon. _____

County of *Lancashire*

Rain Gauge, by _____

Observer _____

J King Jr Esq

- Mersey. Upper H

Diameter	5											
Height	13											
Altitude	110 T											
Year	1890	1891	1892	1893	1894	1895	1896	1897	1898	1899	MEANS.	
January	3.15	2.20	2.82	1.41	2.18	2.41	1.03	1.39	1.67	3.87	2.213	
February	.48	.21	2.10	3.14	3.87	.28	1.66	2.13	2.04	1.05	1.696	
March	2.63	1.44	.94	.46	2.41	2.72	3.84	2.08	1.17	1.69	1.925	
April	.58	1.94	1.26	.68	1.72	1.70	1.61	2.33	1.86	2.44	1.612	
May	2.11	2.26	3.04	1.80	2.20	.58	.54	1.80	3.06	2.51	1.990	
June	3.30	2.32	3.66	1.31	2.98	2.25	2.81	3.48	3.17	1.67	2.695	
July	2.79	2.24	3.47	2.83	2.76	5.45	1.64	1.73	.67	2.31	2.589	
August	3.78	5.50	4.16	2.14	3.98	2.85	2.50	3.48	5.82	1.40	2.561	
September	.84	2.90	3.61	4.13	.93	1.38	5.16	3.82	.60	5.03	2.840	
October	2.06	4.62	4.10	2.34	3.26	3.13	4.10	1.88	3.78	3.16	3.243	
November	4.41	2.22	2.32	2.08	1.70	2.52	1.22	4.80	1.84	1.20	2.431	
December	.37	4.80	2.09	3.28	2.54	3.32	3.55	3.28	3.68	2.22	2.913	
TOTALS	26.50	32.65	33.57	25.60	30.53	28.59	29.66	32.20	29.36	28.55	29.721	

1890

1891

1892

RAINFALL AT Manchester Withington No. 3

Lat. _____

Lon. _____

 County of Lancashire

Rain Gauge, by _____

Observer _____

J. King Jr Esq
Bessey - Upper 14

Diameter	8											
Height	16											
Altitude	110 T											
Year	1890	1891	1892	1893	1894	1895	1896	1897	1898	1899	MEANS.	
January	3.15	2.14	2.81	1.39	2.19	2.39	.99	1.34	1.65	3.73	2.198	
February	.48	.22	2.12	3.06	3.79	.28	1.60	2.07	2.07	1.03	1.672	
March	2.60	1.44	.91	.48	2.40	2.72	3.79	2.08	1.17	1.65	1.924	
April	.59	1.92	1.24	.70	1.71	1.67	1.60	2.31	1.85	2.43	1.602	
May	2.06	2.21	3.00	1.77	2.19	.58	.54	1.80	3.07	2.47	1.969	
June	3.25	2.29	3.58	1.30	2.93	2.24	2.75	3.47	3.14	1.65	2.660	
July	2.78	2.22	3.39	2.79	2.68	5.28	1.60	1.73	.70	2.28	2.545	
August	3.72	5.38	4.09	2.13	3.92	2.81	2.47	3.45	5.74	1.40	2.511	
September....	.83	2.93	3.51	3.99	.94	1.39	5.05	3.79	.58	4.94	2.795	
October	2.04	4.56	4.05	2.32	3.21	3.10	4.06	1.87	3.73	3.14	3.208	
November....	4.41	2.21	2.28	1.99	1.71	2.47	1.20	4.74	1.82	1.14	2.397	
December	.38	4.74	2.06	3.29	2.51	3.37	3.41	3.22	3.59	2.09	2.866	
TOTALS	26.29	32.26	33.04	25.21	30.18	28.30	29.06	31.87	29.11	27.95	27.327	

RAINFALL AT *Manchester Withington, 1891*
Palatine Road
 Lat. Lon. County of *Lancashire South*
 Rain Gauge, by Observer *J. King jr Esq*

Mersey Upper H

Diameter ..	5										
Height	1 3										
Altitude ..	110 T										
Year	1900	1901	1902	1903	1904	1905	1906	1907	1908	1909	MEANS.
January ..	4.11	1.73	2.51	2.54	2.48	1.97					
February ..	2.54	1.39	.95	2.41	2.62	1.53					
March	.67	2.01	1.61	3.43	1.83	3.14					
April	2.12	2.16	1.92	1.60	2.04	2.39					
May	1.52	.85	4.33	2.99	2.12	.66					
June	3.39	1.67	1.07	1.39	.71	2.51					
July	2.79	3.56	1.87	3.02	1.27	2.42					
August	4.24	1.42	3.33	3.58	4.16	4.95					
September ..	1.01	.68	.83	4.54	1.32	2.18					
October	3.73	3.01	3.68	6.10	1.12	2.52					
November ..	1.81	3.44	1.54	3.24	2.19	2.91					
December ..	3.78	4.44	2.27	1.01	2.49	.47					
TOTALS ..	31.71	26.36	25.91	35.85	24.35	27.65					

RAINFALL AT *Manchester* *Walgrave Road* *Widlington, No 2*

Lat.

Lon.

County of *Lancashire*

Rain Gauge, by

Observer *J. King & Co*
L. G. King & Co

Is. Mercury Upper 4

Diameter ..	8										
Height	1.6										
Altitude ..	110 T										
Year	1900	1901	1902	1903	1904	1905	1906	1907	1908	1909	MEANS.
January ..	3.74	1.68	2.54	2.57	2.51	1.97	3.86	1.33	2.13	1.60	2393
February ..	2.50	1.36	.92	2.43	2.59	1.52	3.17	1.71	2.34	1.16	21020
March	.63	1.97	1.65	3.49	1.80	3.20	2.52	2.61	2.40	2.56	21283
April	1.95	2.14	1.93	1.61	2.05	2.38	1.50	1.11	1.94	2.92	1.953
May	1.41	.83	4.36	2.94	2.13	.67	2.60	3.86	2.18	1.53	21251
June	3.10	1.64	1.07	1.38	.72	2.53	1.20	4.85	2.24	1.92	21065
July	2.55	3.44	1.87	2.98	1.26	2.42	1.89	3.05	4.66	4.73	21885
August	3.85	1.41	3.29	3.61	4.16	4.98	2.54	3.36	3.06	2.86	33124
September ..	.96	.71	.82	4.52	1.31	2.18	1.85	.73	3.02	2.07	1.817
October	3.38	2.94	3.73	6.06	1.14	2.53	5.64	3.07	1.49	5.17	3515
November ..	1.62	3.38	1.60	3.22	2.22	2.99	2.28	2.33	2.55	1.10	21329
December ..	3.52	4.36	2.25	1.02	2.49	.45	3.64	2.57	2.48	6.26	21904
TOTALS ..	29.21	25.86	26.03	35.83	24.38	27.82	32.69	30.68	30.99	33.88	31.27

thick

RAINFALL at

Wilmington Pelatine Road

County of South Lancashire

Observer L. G. Kingdon

River Basin Upper Mersey 4

Rain Gauge, by

Diameter ...	8											
Height	1 6											
Altitude ...	110 ft											
Year ...	1910	1911	1912	1913	1914	1915	1916	1917	1918	1919	MEANS.	
January ...	3.20											
February ...	2.55											
March	#											
April												
May												
June												
July												
August												
September ..												
October ...												
November ..												
December ...												
TOTALS ...												

Date.	Lat. N.	Long.	Railway Station.	Distance.	Direction.	Church.	Distance.	Direction.